Environmental Racism and AI Data Centers*

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Abstract

This study analyzes the structural drivers and disparate geographic impacts of artificial intelligence (AI) data centers through a spatial political economy perspective. We conceptualize AI infrastructure through a spatial dichotomy: Back-end Training Centers and Front-end Inference Centers. We argue that back-end facilities, driven by cost-minimization and historical path dependence (e.g., redlining), disproportionately cluster in vulnerable communities to create modern "sacrifice zones" of concentrated ecological externalities. Conversely, front-end inference centers—requiring low-latency proximity to urban cores—act as catalysts for tech-gentrification. Utilizing historical analogies to Urban Green Spaces (UGS) and empirical cases such as Loudoun County, Virginia, we demonstrate how these urban facilities trigger localized climate degradation, "tax illusions,"increase in living costs, and the displacement of working-class populations. By mapping these systemic causal chains, this paper establishes critical criteria for future spatial governance, emphasizing the internalization of back-end ecological costs and the mandate of front-end value capture mechanisms to ensure a spatially equitable AI transition. This paper presents the theoretical framework and preliminary hypotheses. Empirical data and full analysis will be included in subsequent revisions.

Keywords: Environmental Justice, Spatial Political Economy, Tech-Gentrification, Artificial Intelligence, Digital Redlining

*This is a preliminary draft. Comments are welcome.

†Department of Sociology, University of Washington. Contact: me@z1w.net. This is a work in progress. While every effort has been made to ensure accuracy, the author assumes no responsibility for any consequences arising from the use of these preliminary findings. Comments are welcome; all remaining errors are my own.

1 Introduction

The rapid ascent of Artificial Intelligence (AI) often presents a clean, invisible utopia of zero-latency services and autonomous efficiency. However, this digital era maintains a profound visual and psychological deception: the ”beautiful vision” of front-end applications systematically masks the brutal physical reality required to sustain them. This study seeks to deconstruct the traditional narrative of AI as a ”clean tech” industry by exposing the systemic spatial dispossession and environmental crises hidden beneath the mask of infrastructure modernization.

We categorize the physical footprint of AI into a geographic and economic dichotomy. Back-end Training Centers, the energy-intensive ”furnaces” of large models, often cluster in remote or marginalized regions to exploit cheap land and lax regulations. Meanwhile, Front-end Inference Centers embed themselves within dense urban cores to achieve the ”low-latency ecological niche” required for real-time services.

Building upon digital infrastructure literature, we identify a significant research gap. While sociological research has documented the impact of back-end centers, the systematic distribution of front-end urban facilities remains under-examined. Drawing on theories of ”NIMBYism” and ”tech-gentrification,” we analyze how back-end centers exacerbate health inequities through ”Computational Extraction,” while front-end facilities catalyze ”tax illusions” that drive economic eviction. By mapping these causal chains, we establish critical criteria for future spatial governance.

2 Literature Review & Framework

Building upon the evolving literature on digital infrastructure, this study integrates perspectives from spatial political economy and urban economics to analyze the bifurcated spatial distribution of AI infrastructure. While existing sociological research has extensively documented the impact of back-end training centers on marginalized communities, there is relatively less attention paid to the ”front-end” urban facilities. Traditionally, industrial siting follows the ”path of least resistance,” where historical structural inequities—such as 20th-century redlining—depress land values and weaken the political capital of minority communities, transforming them into modern ”sacrifice zones.” However, the AI revolution is fundamentally a dual spatial process. As Avelar et al. (2023:2) demonstrate in Table 1, inference workloads significantly exceed those of training, necessitating a shift in focus toward ”front-end” inference centers. Unlike remote back-end models, these facilities must be physically embedded within dense urban cores to provide low-latency services like autonomous driving and algorithmic trading. Drawing interdisciplinary inspiration from ”tech-gentrification” and the uneven distribution of urban green spaces (UGS), this paper explores

beyond the pollution of back-end training centers and analyzes how front-end facilities trigger asset appreciation, rising living cost, and ultimately catalyzing the economic eviction of the urban working class.

3 Back-End Training centers – "Sacrifice Zones"

Back-end training centers are the "furnaces" of AI large models. The core objective is to sustain computational clusters regardless of resource (like water and electricity) constraints. To minimize operational costs, tech giants target vulnerable regions with lax regulations and cheap resources. Unlike the traditional "persecution conspiracy" narrative, this study focuses on analyzing various factors that "attract" tech giants. Driven by profit maximization, these decisions conform to systemic, structural inequalities. Historically, these areas were heavy industrial zones, and due to legacy issues like 20th-century "redlining," minority residential areas were confined to the peripheries of these industrial belts where land and power costs are cheap. Moreover, the strong "Not In My Back Yard" (NIMBY) effect in affluent neighborhoods creates significant hurdles for site selection, naturally making minority communities with weaker political agency the "path of least resistance." All these structural factors perfectly align to facilitate the natural settlement of data centers. However, setting subjective motives aside, these purely cost-and-efficiency-driven business decisions have objectively caused severe devastation to environmental justice, and tech companies must still be responsible for resulting systemic racist consequences.

"Sacrifice Zones" originally referred to specific geographical areas bearing the pollution of heavy chemical industries. In the context of AI infrastructure expansion, as Booker (2025) points out, this concept acquires a new dimension: it specifically refers to "communities systematically designated as acceptable locations for environmental harm in service of broader economic interests." This siting strategy is not coincidental but a modern replication of the discriminatory 1930s housing policy of "redlining," forming a "Digital Redlining." Macro-siting data indicate that training centers are highly concentrated in the U.S. South and Midwest. In historically Black communities like South Memphis, Tennessee, or Louisiana's industrial belts, AI data centers are gradually replacing traditional energy-intensive industries, reducing these areas into "sacrifice zones" where environmental harm has become normalized.

The physical training process of AI large models is accompanied by highly significant environmental externalities. Take Elon Musk's xAI supercomputer project in Memphis as a case study. The host community, South Memphis, has a 64% Black population, a median income less than half the national average, and was historically redlined for federal loans (Booker, 2025). The area is already surrounded by 17 toxic waste sites, and residents face cancer risks four times higher than the national average. Following this "path of least resistance," xAI deployed and operated

35 gas turbine generators without securing proper environmental permits to compensate for the massive power deficit in the local grid. Environmental assessments indicate that these units emit formaldehyde, nitrogen oxides, and PM2.5 into an already vulnerable community. Their massive cooling systems consume millions of gallons of industrial water, accompanied by constant 24/7 low-frequency noise pollution. Furthermore, regional energy studies by authoritative think tanks such as the Brookings Institution (Perry, 2024) show that due to the scaled crowding-out of grid base loads by data centers, residential electricity prices in some host regions have experienced significant abnormal increases. Tech giants bypassing normal environmental reviews to fast-track construction objectively exacerbates the health inequities and cumulative exposure burdens already faced by marginalized communities.

Local governments often anticipate scaled employment dividends when introducing such projects. However, training centers possess an extremely high organic composition of capital. Statistical evaluations indicate that every $100 million in infrastructure investment translates into very few long-term local maintenance jobs, mostly limited to basic security and cleaning roles (Urban Institute, 2024; Gross et al., 2005). This economic model is termed "Computational Extraction": tech enterprises utilize the cheap land, water, and electricity resources of vulnerable communities to extract computational power for the global market, while the communities themselves struggle to share in core technological dividends or achieve industrial cluster upgrades, instead unilaterally bearing the majority of negative externalities.

4 Front-End Inference Centers and Insidious Spatial Eviction

Front-end Inference Centers are responsible for processing immediate user requests (e.g., AI assistance/conversations, autonomous driving) and must be located near densely populated commercial districts to achieve a "low-latency ecological niche." However, beneath the mask of "infrastructure modernization" lies a systemic spatial eviction of marginalized communities.

4.1 Asset Appreciation and Spatial Dispossession: Historical Analogy Based on UGS

To understand the spatial dynamics of front-end centers, the historical development of Urban Green Spaces (UGS) provides a compelling analogy for how infrastructure can drive asset appreciation rather than public welfare.

In the history of rapid urbanization in emerging economies, urban development is often chaotic and market-driven. Early planning prioritized economic centers, leading to a highly uneven geographical distribution of infrastructure, notably Urban Green Spaces (UGS). Quantitative research

by Bressane et al. (2024) indicates that green space was frequently alienated as a "tool for asset appreciation" rather than distributed for "public welfare." Historical social structures pushed marginalized groups into under-invested "green deserts," while capital accumulation enabled high-income groups to occupy premium green lands, driving up property values to form exclusive "green fortresses." Ultimately, UGS evolved into a physical carrier of environmental racism defined by its systemic "absence" in minority communities.

As the core infrastructure of the 21st century, AI inference centers exhibit striking logical similarities to UGS in terms of capital accumulation. They are not merely technological hubs but new markers of land premium and resource exclusivity. As seen in San Jose, California, the establishment of inference centers is typically accompanied by high-specification power grid upgrades and fiber-optic deployment. The surrounding land undergoes a similar transformation, becoming a new tool for asset appreciation. Although AI data centers require cheap land and electricity, their indispensable need for high-speed backbone networks and stable power supply leads them to "invade" low-income, marginalized areas that possess favorable geography (Booker, 2025). This modernization significantly drives up local land values, leaving less space for marginalized groups to live and effectively driving indigenous residents away.

4.2 Climate Accelerators and Health Crises: UHI

If historical urban green spaces practiced inequality through the "absence" of premium resources, contemporary AI data centers perpetuate this logic through the "occupation and transformation" of physical spaces. This occupation manifests not only in land prices but also in the severe deterioration of local climates. Inference centers generate immense waste heat during operation, which is exhausted directly into surrounding communities, significantly exacerbating the Urban Heat Island (UHI) effect. AI data centers are not isolated technological entities but accelerators of localized climate crises. According to Majeed and Lee (2017), the impacts of climate change and extreme heat are disproportionately larger on vulnerable populations, especially causing severe health damage to minority youth. By materializing global computational demands into localized extreme heat crises, inference centers force people into shrinking habitats and deteriorating physical and mental health within an increasingly sweltering urban jungle.

4.3 Tax Illusion and Spatial Devouring: Loudoun County

Local governments are often eager to introduce front-end inference centers and hybrid data hubs in pursuit of substantial property taxes. However, as demonstrated by Loudoun County, Virginia—the central hub of the global internet—this prosperity driven by capital-intensive infrastructure ultimately evolves into a profound "Tax-Job Paradox" and the spatial devouring of low-income groups.

On the surface, data centers contribute staggering financial revenues to Loudoun County. According to the county's fiscal year budget report (Loudoun County Government, 2024), tax revenues generated by data centers account for a vast majority of the county's personal property taxes. This allowed the local government to reduce the nominal real estate tax rate from $1.285 to $0.805 per $100 in assessed value over a decade. However, this triggers a cruel "Tax Illusion": although the nominal tax rate decreased, the absolute land valuation skyrocketed due to the presence of data centers. Consequently, the actual property taxes and rent expenses faced by ordinary residents have surged continuously, rendering the local blue-collar working class completely unable to afford the cost of living.

Inference centers increasingly monopolize the city center's limited high-quality land, clean electricity, and water resources. An authoritative report by the Virginia Joint Legislative Audit and Review Commission (JLARC, 2023) points out that tech giants, in their quest to expand computational power, often outbid residential developers by purchasing land at exorbitant prices of one to two million dollars per acre. According to an assessment by the George Mason University Center for Regional Analysis (GMU CRA, 2024), this land devouring has directly contributed to a regional housing shortage of over 75,000 units in Northern Virginia. Land that could have been used for affordable housing or minority commercial complexes has been transformed into cold, extremely low-employment-density server farms. These massive infrastructures typically offer very few maintenance jobs, forming an invisible "Digital Redlining" against local residents. Furthermore, when ordinary low-income communities attempt to upgrade their power grids, they often despairingly find that regional power quotas have been entirely consumed by the adjacent "AI brains."

5 Discussion and Policy Implications

To mitigate the severe environmental footprint of back-end centers, we propose transitioning from voluntary corporate social responsibility (CSR) to a uniform legal incentive framework. While Google DeepMind has demonstrated that AI-driven cooling optimization can reduce energy consumption by up to 40% (Varghese 2022), such practices must be codified as legal obligations. Governments should implement emission targets and water/electricity efficiency standards, requiring data centers to achieve these targets as a licensing prerequisite. To break the "path of least resistance" in siting, policy frameworks must force the internalization of environmental costs. We recommend dynamic carbon taxing and tiered water pricing, ensuring that the pursuit of computational power remains financially tethered to ecological limits.

Regarding front-end facilities—which drive urban asset appreciation—policy must address the "Loudoun County Paradox," where increased tax revenue often fails to offset rising local living

costs (e.g., surging electricity rates). We recommend implementing mandatory Value Capture mechanisms, where a fixed percentage of data center tax revenue is legally ring-fenced for affordable housing trusts and local energy subsidies.

For both facility types, we propose a "Community-Coupled Infrastructure" mandate. Tech giants' upgrades to high-speed fiber optics, stable power grids, and water systems are marginal costs compared to their total capital expenditure. Legislation should require that any infrastructure upgrade for a data center must simultaneously enhance the local community's utility access. Agreements must stipulate a mandatory quota for local hiring, coupled with company-funded technical training programs.

6 Conclusion

The prosperity of the AI era is currently constructed upon a cruel, bifurcated logic: while the global market enjoys high-end digital services, marginalized communities bear the weight of factory-like pollution on the back-end and suffer spatial dispossession on the front-end. This study has demonstrated that back-end "sacrifice zones" are not coincidental; they are modern replications of 1930s "redlining" policy, now manifesting as Digital Redlining where environmental harm is normalized in the name of economic interest. Simultaneously, front-end facilities in regions like Loudoun County reveal a "Tax-Job Paradox," where nominal tax reductions are overshadowed by skyrocketing land valuations and a regional housing shortage.

To counter this dual environmental racism, traditional financial subsidies are insufficient. We propose a transition from voluntary corporate social responsibility to a uniform legal incentive framework. For the back-end, this includes codifying AI-driven energy optimization—such as the 40% efficiency gains seen in Google DeepMind—into mandatory legal obligations and tiered water pricing. For the front-end, we advocate for Value Capture mechanisms where tax revenues are legally ring-fenced for affordable housing and community energy subsidies.

Ultimately, AI infrastructure must be governed by a "Community-Coupled Infrastructure" mandate. Any technological upgrade to power grids or fiber optics must simultaneously enhance local utility access and human capital through local hiring quotas. Only by rectifying these hidden spatial costs can AI be transformed from a tool of displacement into a genuine public good that supports, rather than devours, the communities it inhabits.

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